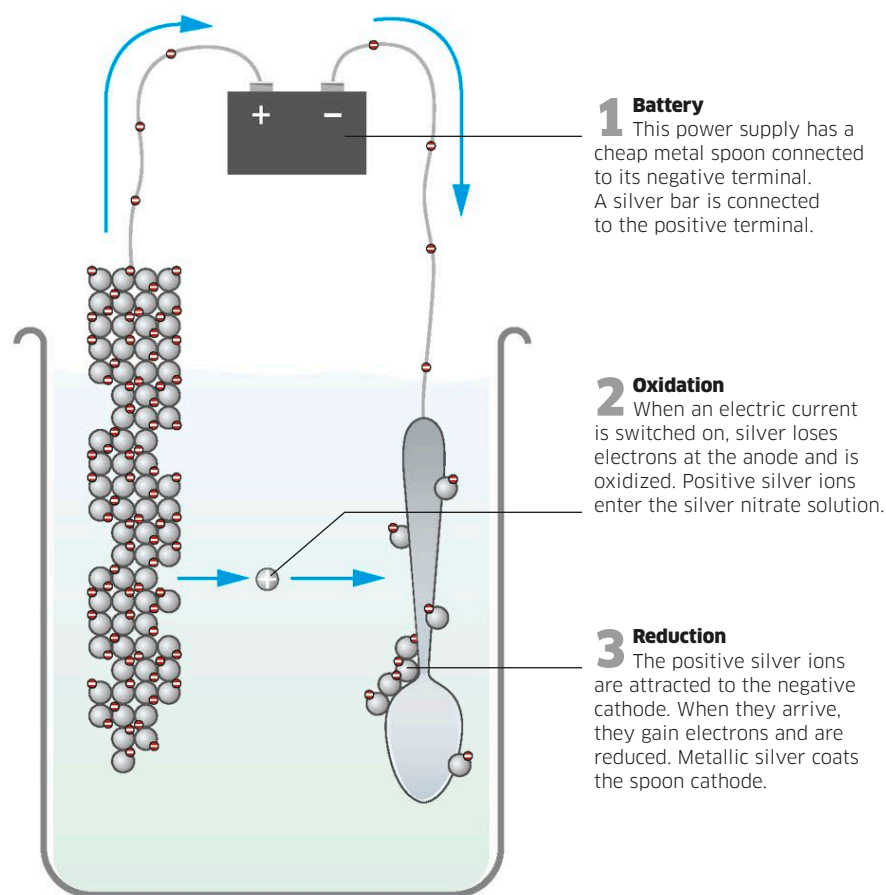


Electroplating

Similar to electrolysis, electroplating is a process that coats a cheaper metal with a more expensive metal, such as silver. To turn a cheap metal spoon into a silver-plated spoon, the cheap metal spoon is used as a cathode (negative electrode) and a silver bar is used as the anode (positive electrode). These two electrodes are bathed in an electrolyte that contains a solution of the expensive metal, in this case silver nitrate solution.



Gray tarnish showing that oxidation has taken place.



Oxidation in air

Silver oxidizes when exposed to air, so silver plated items eventually lose their shine as a gray tarnish forms on the surface. Polishing removes the tarnish but the plating might be damaged.

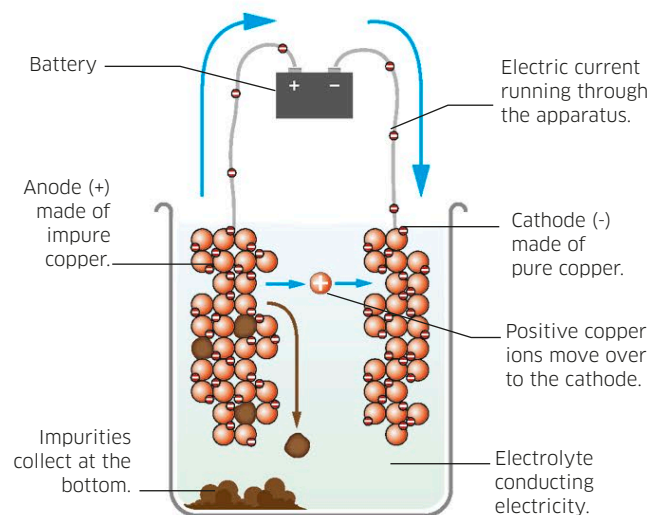
Galvanizing

Steel or iron can be prevented from rusting (a form of oxidation) by coating them in the metal zinc, a process called galvanizing. These nails have been galvanized.



Purifying metals

The copper that is extracted from copper ore is not pure enough to become electrical wiring. It has to be purified by electrolysis. Impure copper acts as the anode, and pure copper as the cathode. These electrodes lie in a solution of copper sulfate.



Electrorefining

Pure copper is used to make electrical wiring and components. Here you can see copper purification, called electrorefining, being carried out on a massive scale in a factory, in the process described above.



Electrochemistry in batteries

Batteries turn chemical energy into electrical energy (see p.92). This is the opposite of electrolysis, which turns electrical energy into chemical energy. In a battery, it is the anode that is negative and the cathode that is positive. The reaction at the anode is still oxidation and at the cathode it is still reduction.

